

Lab Project

Final Report

September 24, 2026

Multi-Object Tracking of Polybags

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Abstract

Multi-object tracking of polybags on a conveyor belt is a core enabling technology for automated parcel sorting. This project develops and evaluates a complete multi-camera tracking pipeline for deformable polybags observed by overlapping top-view cameras, pursued along two complementary tracks. The first track builds a fully controllable synthetic environment: a physics-based simulation of polybags on a conveyor is rendered in Blender from four cameras, and a geometry-driven annotation pipeline produces pixel-accurate oriented-bounding-box labels and tracking ground truth without any manual labelling. A YOLO detector trained on this data reaches a mean average precision (mAP50) of 0.995, and single-camera trackers (ByteTrack and BoT-SORT) together with several multi-camera association strategies are benchmarked on standard tracking metrics, reaching a multi-object tracking accuracy of up to 97.9%, an IDF1 of up to 92.6%, and a HOTA of up to 86.9. The second track, which is the focus of the project, transfers this pipeline to the physical rig. The cameras are calibrated onto a shared metric belt plane, and real conveyor footage is auto-labelled with a fine-tuned vision-language model and used to train a detector. A live digital twin then projects every detected bag onto a two-dimensional map of the belt in real time over a lightweight message bus, where detections are fused across cameras and counted by occupancy flux. This report describes the methods, results, and limitations of both tracks, and the lessons learned in moving from a synthetic benchmark to a real conveyor system.

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1 Introduction

The courier, express, and parcel industry has seen sustained growth in shipment volumes, which raises the demand on automated sorting systems. A large and growing share of these shipments are polybags: soft, flexible plastic packages that do not hold a fixed shape. Unlike rigid boxes, polybags deform, overlap, and occlude one another on a moving belt, and they look very different from one viewpoint to the next. This makes them hard to detect, track, and handle with conveyor systems built for rigid parcels.

Recent advances in computer vision and deep learning have made vision-based perception on a conveyor practical. Modern object detectors locate items reliably in cluttered scenes, and tracking-by-detection methods such as ByteTrack [15] and BoT-SORT [1] maintain object identities across a video stream. A single camera, however, sees only part of a wide conveyor and loses items to occlusion. Covering the belt requires several cameras with overlapping fields of view. That in turn raises the problem of recognising that a bag seen by one camera is the same bag seen by another.

The perception problem therefore decomposes into three levels of increasing difficulty. The first is detection: separating each polybag from the background and from its neighbours. The second is detection with tracking: following each bag across a sequence of frames while preserving a consistent identity. The third is detection with tracking and cross-camera association: keeping that identity consistent not only over time but also across the multiple cameras that observe the belt. Unstable identities, seen as identity switching or flickering between frames or cameras, translate directly into sorting errors downstream. That is why robust multi-camera tracking is the central concern of this work.

This project addresses the problem along two complementary tracks: a synthetic track builds the full pipeline in a controllable simulation with exact ground truth, and a real track, the focus of the project, transfers it to the physical conveyor rig and its recorded footage.

2 Task Description and Project Goals

The overall task is to design and evaluate a system that tracks multiple deformable polybags across multiple camera views on a conveyor belt in real time, while preserving a consistent identity for each bag across both frames and cameras. Beyond identity, the system should recover the information a downstream actuator would need for automated sorting, in particular the position of each bag on the belt and, where possible, physical properties such as size and shape.

Three families of quantitative goals guide the work. Detection quality is measured by mean average precision (mAP). Tracking quality is measured with the established CLEAR MOT metrics [3], chiefly the multi-object tracking accuracy (MOTA) and the number of identity switches, together with the identity-preservation measure IDF1 [12]. Real-time behaviour is

measured by throughput in frames per second, since a sorting system must keep pace with the belt.

The task is pursued along two tracks that share one pipeline. The synthetic track renders polybags on a conveyor in Blender and generates its own ground truth automatically, giving a perfectly labelled and fully repeatable setting in which to build and benchmark every component. The real track applies the same pipeline to the physical rig, where the cameras are not synchronised, ground truth is scarce, and the bags no longer separate neatly by colour. The real conveyor system is the ultimate target of the project, and the synthetic track serves as the controlled proving ground that makes building it tractable.

Reaching these goals requires several intermediate steps: generating the synthetic dataset and its annotations, training an object detector, adding single-camera tracking, extending tracking to cross-camera association, calibrating the real rig to a common metric frame, and finally combining detection, tracking, and calibration into a live digital twin of the belt.

3 Scientific/Technical Status at Project Start

At the start of the project the physical conveyor rig was available in the Conveyor and Material Technology Laboratory, instrumented with five cameras that can record together: two Basler GigE cameras, one Lucid Triton GigE camera, and two Intel RealSense D435 depth cameras, each contributing a colour and a depth stream. None of the cameras had yet been calibrated. Recorded footage of polybags on the belt was to be provided by the supervisor rather than captured by the team at the outset, so the early work could proceed without physical access to the rig.

On the software side, the team could build on mature open-source components: the Blender simulation and rendering environment for the synthetic track, the Ultralytics YOLO family of detectors [7] with publicly available pretrained weights, and reference implementations of the tracking-by-detection methods. Geometry describing the polybags was available to drive the simulation. No project-specific detection or tracking code existed at the start. The pipeline was to be assembled and extended from these components.

3.1 Related work

A short literature review framed the design choices and is summarised here.

Detection. Modern one-stage detectors such as the YOLO family [7] locate objects reliably in cluttered scenes. Oriented bounding boxes suit elongated, rotated objects like polybags on a belt, where an axis-aligned box wastes area and merges neighbours. Small or densely packed objects remain hard, which motivates tiled inference schemes such as SAHI [2].

Tracking. Multi-object tracking is dominated by the tracking-by-detection paradigm, in which per-frame detections are linked over time. DeepSORT [14] adds an appearance embedding

to a motion model. ByteTrack [15] instead keeps low-confidence detections and matches by overlap, and BoT-SORT [1] adds camera-motion compensation and optional re-identification. Transformer-based trackers such as TrackFormer [11] learn association end to end. Performance is assessed with the CLEAR MOT metrics [3] and IDF1 [12], the de facto standards.

Multi-camera association. Extending tracking to a shared identity space across cameras follows two broad strategies. Appearance-based re-identification matches objects by learned features. Calibration-free heuristics use cheap cues such as colour class, left-to-right ordering, and temporal overlap. Both assume either distinctive appearance or overlapping, synchronised views. Geometry-based association instead maps every camera onto a common ground plane through calibration [16, 5] and matches objects by their metric position. Which strategy is viable depends entirely on the data, a dependence that turns out to decide the approach taken on the real rig (Section 5.2).

4 Project Plan

The project was planned in four phases running from late April to late September 2026. The timeline drawn up at the kickoff is reproduced in Figure 1, and the phases and their milestones are described below.

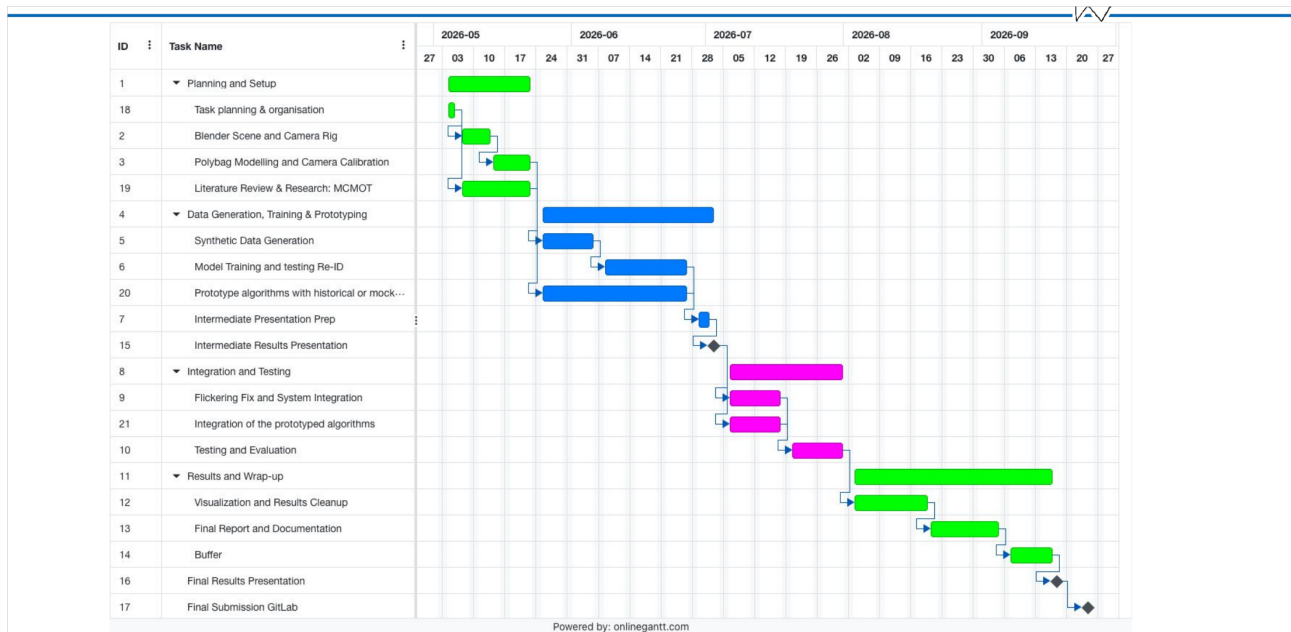


Figure 1: Project timeline and milestones as planned at the kickoff.

The first phase, *Planning and Setup* (May), covered task planning and organisation, building the Blender scene and camera rig, modelling the polybags and calibrating the cameras, and a literature review of multi-camera tracking. The second phase, *Data Generation, Training and Prototyping* (June), covered synthetic data generation, detector training and re-identification experiments, and prototyping the tracking algorithms on mock data. It closed with the intermediate presentation, the first major milestone, held on 2 July 2026.

The third phase, *Integration and Testing* (July to early August), covered fixing identity flickering, integrating the prototyped algorithms into a single system, and testing and evaluating it, now on real conveyor data. The fourth phase, *Results and Wrap-up* (August to September), covered visualisation and results cleanup, the final report and documentation, a scheduling buffer, the final results presentation, and the final submission. The principal milestones were therefore the intermediate presentation, the final results presentation, and the final submission.

5 Project Execution and Scientific/Technical Results

The project was carried out along the two tracks introduced above. The synthetic track came first and formed the intermediate deliverable. The real track, which is the focus of the project, then carried the same ideas onto the physical conveyor. The two are presented in that order.

5.1 Synthetic Track

The synthetic track builds and benchmarks the whole pipeline on a simulated conveyor, where the ground truth is exact and every experiment can be repeated. That is why it was built first.

5.1.1 Data generation and automatic annotation

Polybags on a conveyor were simulated with a physics engine and rendered in Blender from four cameras (front, back, left, and right) at 1920 by 1080, 1841 frames in seven colour classes (Figure 2). The simulation knows each bag's exact geometry, so both label sets are generated automatically, with no manual work (Figure 3). The oriented boxes come from projecting each mesh and fitting a minimum-area rectangle, and the colour class is read at the projected centroid. The tracking ground truth comes from Hungarian matching of the three-dimensional centroids frame to frame, which gives a stable identity per bag (66 341 rows). Projection is pixel-accurate, lighting-insensitive, and occlusion-aware by construction. Manual labelling and a colour-threshold auto-labeller are not (Table 1).

Both label sets come from a single headless Blender script, `blender_annotate.py`. It imports each frame's STL geometry, separates the loose parts into individual bags, projects every mesh vertex through the camera matrix (`world_to_camera_view`), and fits a minimum-area rectangle to the projected hull, writing an eight-coordinate oriented box per bag per camera. A second

pass, `relabel_synth.py`, reads the colour class from a 13×13 pixel patch at each box centroid and fixes it by a per-track majority vote across all frames. This pass also surfaced a seventh class, teal, that the raw material indices had not separated. The geometry is exact, so the boxes are mathematically correct and independent of lighting, colour, or occlusion.

Producing a usable multi-camera set took one round of replanning worth recording. The first render ran in two batches, and each was interrupted, so every camera ended up with a different set of missing frames. Only 35 frames existed in all four cameras at once. That is far too few for cross-camera evaluation, because a tracker fragments a track at every gap and the resulting scores are meaningless. The sequences were re-rendered with four parallel Blender instances each, using a skip-logic that never reprocesses a completed frame. After that, every frame is present in every camera.

Table 1: Why geometric projection was chosen over the alternatives for the synthetic labels.

Criterion	Manual	HSV auto-labeller	Blender projection
Accuracy	high (if careful)	approximate	pixel-perfect
Lighting sensitivity	none	high	none
Occlusion handling	manual	fails	geometry-based
Scales with frames	no	yes	yes
Manual effort	high	tuning required	zero

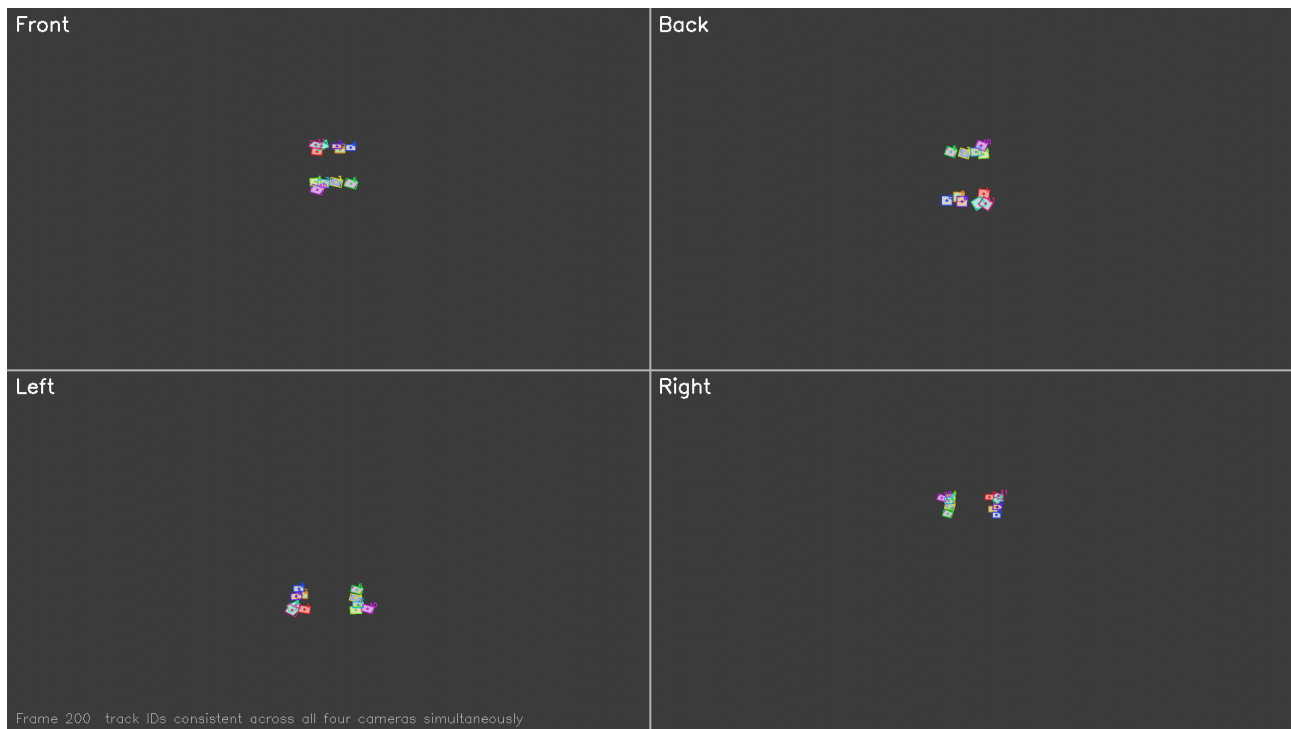


Figure 2: Synthetic scene rendered from the four cameras (front, back, left, right).

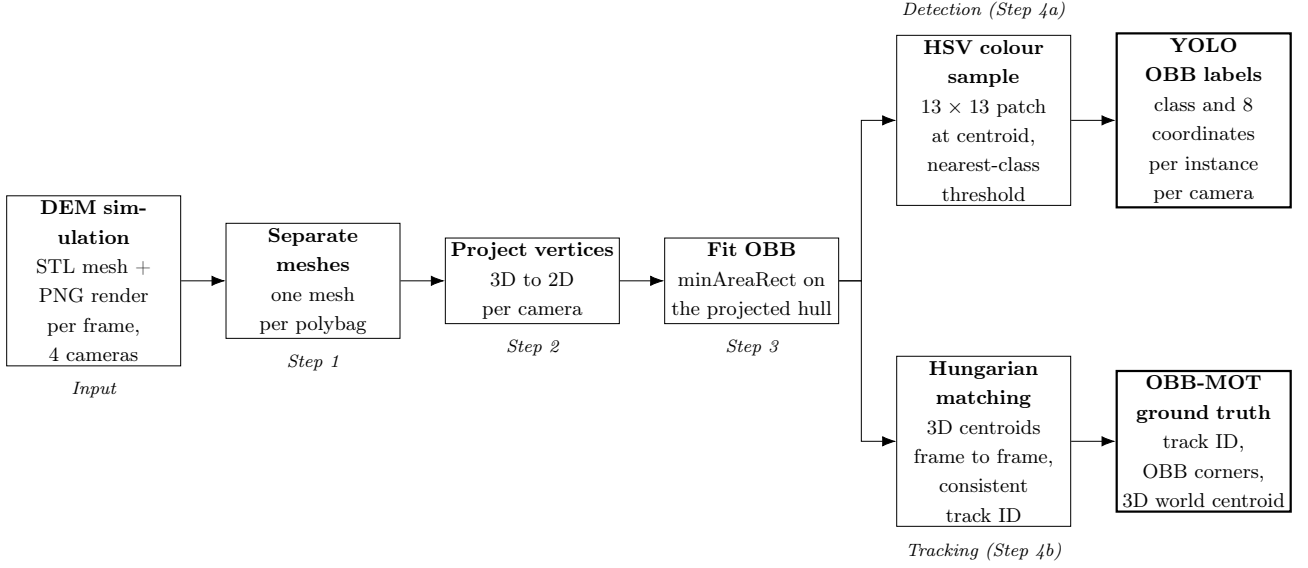


Figure 3: Automatic annotation pipeline: mesh projection to oriented boxes (Step 4a) and Hungarian centroid matching to tracking ground truth (Step 4b).

5.1.2 Dataset

The frames were split by time into disjoint ranges (Table 2), so no frame next to a training frame leaks into evaluation.

The dataset carries seven colour classes and eleven particle identities, and these stay globally consistent across every camera and split (Figure 3, Step 4b). The identities come from Hungarian matching of the three-dimensional world centroids between consecutive frames. A particle within 0.30 Blender units of a previous one keeps its identity; otherwise it starts a new one. All four cameras share the same geometry for a given frame, so the same assignment applies to all of them at once. Each camera sequence is written twice: as `gt_obb.txt`, an extended sixteen-field record (frame, identity, the four oriented-box corners, confidence, class, visibility, and the three-dimensional world centroid), and as a standard MOT16 `gt.txt` for legacy tools, with a MOTChallenge `seqinfo.ini` descriptor. The seven classes are unevenly populated. In the training split yellow is the most common (6447 instances) and green the least (1686). The gaps between splits are deliberate too (frames 600 to 999 and 1251 to 1499 are left out), so the validation and test states are configurations the detector never saw.

Table 2: Synthetic dataset splits.

Split	Frame range	Frames	Images	YOLO ann.	MOT rows
Train	100 to 599	500	2000	22 000	22 000
Val	1000 to 1250	251	1004	7584	8584
Test	1500 to 1750	251	1004	6024	7024
Total		1002	4008	35 608	37 608

5.1.3 Detection

An oriented YOLO detector (YOLO11-ORB, Ultralytics [7]) was trained on the OVGU cluster (NVIDIA A40) at input size 1920. On the synthetic test set it reaches mAP₅₀ 0.995 and mAP (0.5 to 0.95) 0.989, a near-ceiling result on this clean, exactly-labelled data.

5.1.4 Tracking and multi-camera association

Two single-camera trackers were evaluated. ByteTrack [15] uses a Kalman filter plus intersection-over-union and keeps low-confidence detections, with no appearance re-identification, which suits same-colour bags. BoT-SORT [1] adds camera-motion compensation and optional re-identification. For cross-camera identity, several associators were used: an offline family as an accuracy ceiling (temporal, spatial, combined) and a causal real-time family (class-rank, class-iou, class-smooth).

5.1.5 Results

BoT-SORT is the best single-camera tracker (MOTA 97.9%, IDF1 92.6%), and ByteTrack is about a third faster (Table 3, Figure 4). The offline associators reach the accuracy ceiling. The real-time ones stay accurate online, at the cost of more identity switches. Along the way, two benchmark defects were found and fixed: an identity collision in the greedy track-merging step, and a detection degradation under sequential multi-camera processing, solved by isolating each camera in its own subprocess.

Table 3: Tracking results on the synthetic test set. Higher is better for MOTA, IDF1, and FPS; lower is better for identity switches. FPS is not applicable to the offline associators.

Category	Method	MOTA	IDF1	ID Sw.	FPS
Single-camera	ByteTrack	95.3%	91.1%	47	27.1
	BoT-SORT	97.9%	92.6%	36	19.7
Offline	trk.temporal	97.8%	72.5%	53	n/a
	trk.spatial	97.8%	52.5%	59	n/a
	trk.combined	97.8%	61.3%	54	n/a
Real-time	class-rank	96.4%	61.7%	342	5.78
	class-iou	97.7%	55.9%	64	5.76
	class-smooth	97.0%	61.5%	212	5.78

HOTA [10] was also requested for the synthetic track, so that its trackers read on the same scale as the real one (Table 8). It was computed with TrackEval on the same 500-frame multi-camera sequence that produced the MOTA and IDF1 figures above. The per-camera MOTA

reproduces the reported values exactly (ByteTrack mean 95.3, BoT-SORT mean 97.9), which confirms these are the same runs and not a re-evaluation. Table 4 gives the decomposition.

Table 4: HOTA on the synthetic track for the two single-camera trackers, per camera, on the 500-frame multi-camera sequence. HOTA with its detection (DetA) and association (AssA) components, alongside MOTA and IDF1. Higher is better; values are percentages.

Tracker	Camera	HOTA	DetA	AssA	MOTA	IDF1
ByteTrack	front	82.8	84.6	81.5	93.4	91.6
	back	91.1	90.4	92.1	98.1	99.0
	left	80.9	83.2	78.8	93.9	88.6
	right	75.5	79.0	72.7	95.8	85.3
	mean	82.6	84.3	81.3	95.3	91.1
BoT-SORT	front	87.0	89.5	85.0	97.8	93.1
	back	93.4	92.8	94.3	98.3	99.0
	left	86.2	89.2	83.5	97.6	90.4
	right	80.8	83.6	78.6	98.0	88.0
	mean	86.9	88.8	85.3	97.9	92.6

HOTA confirms the MOTA and IDF1 ranking. BoT-SORT is the stronger single-camera tracker (mean HOTA 86.9 against 82.6), and the four-point gap comes from both components, a higher DetA (88.8 against 84.3) and a higher AssA (85.3 against 81.3). Thus its camera-motion compensation and appearance cues help detection and identity linking alike. Both trackers sit far above the real track (the greedy tracker’s HOTA of about 66). This quantifies how much the clean simulation, the single re-render at full frame coverage, and the seven-colour cue simplify tracking. The decomposition also differs in character. The real track is association-limited (high DetA, low AssA); here DetA and AssA are both high and close, so neither detection nor association dominates the residual error. The per-camera spread is the expected one, and it is consistent across both trackers: the back view is the easiest (HOTA 91.1 to 93.4) and the right view the hardest (HOTA 75.5 to 80.8), which tracks the viewpoint difficulty already visible in the detection figures.

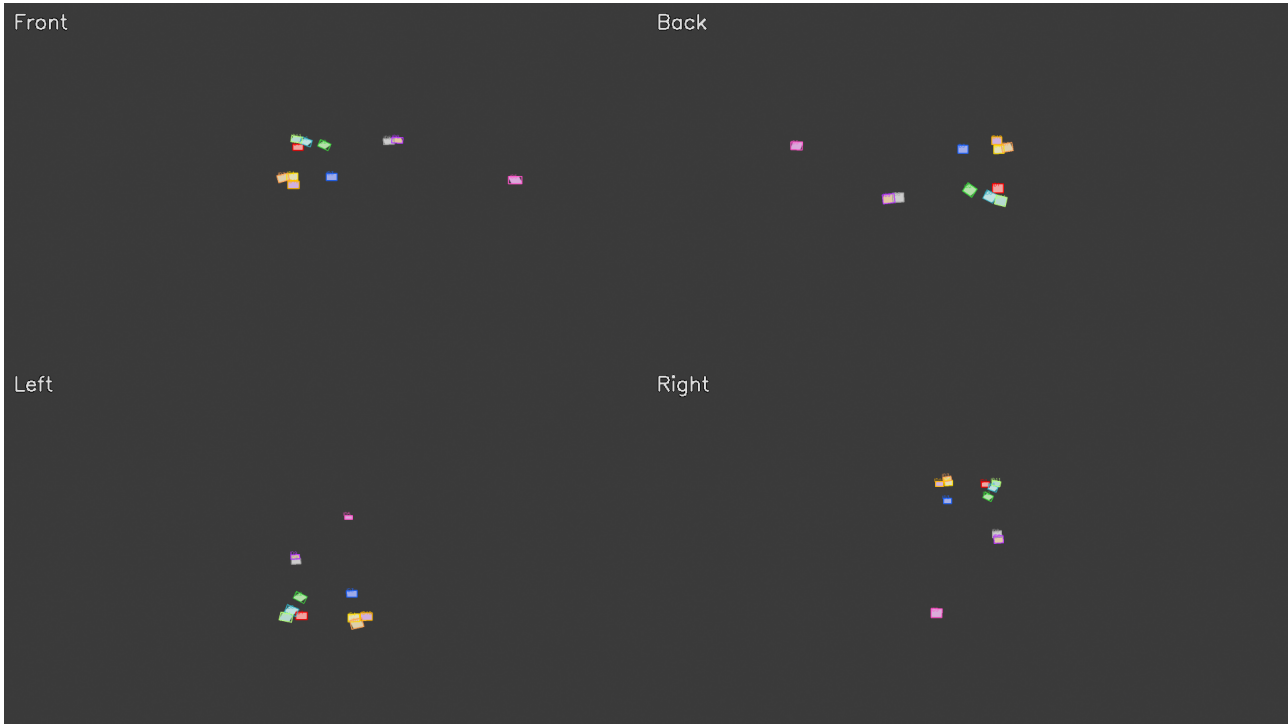


Figure 4: Tracked polybags on the synthetic footage across the four cameras, with per-object identities.

5.2 Real Track

The real track carries the pipeline onto the physical rig and is the focus of the project. The rig carries five cameras (two Basler GigE, one Lucid Triton, two Intel RealSense D435, each colour plus depth) recording the belt at 1280 by 720. The belt is one-directional and 580 mm wide (Figure 5). Two facts make the real data far harder than the simulation and shape everything below: the bags are a single merged class (no colour cue), and the moving recordings have no timestamps, so the cameras are not truly synchronised. Both break the synthetic association cues and motivate one shared metric belt frame, in which association becomes a distance in millimetres.

5.2.1 Why a calibrated metric frame

The synthetic track associated cameras with calibration-free cues: the seven colour classes and a left-to-right ordering under synchronised views. On the real rig both cues collapse. The bags are a single merged class, so colour and appearance carry no identity, which rules out the appearance-based and colour-gated strategies from the literature. And the moving footage is not synchronised, so ordering and temporal-overlap heuristics have no common clock to stand on. The one remaining sound basis is geometry: map every camera onto the same metric belt plane and associate by position in millimetres. This is why the project committed to calibration rather than to a heuristic association layer.

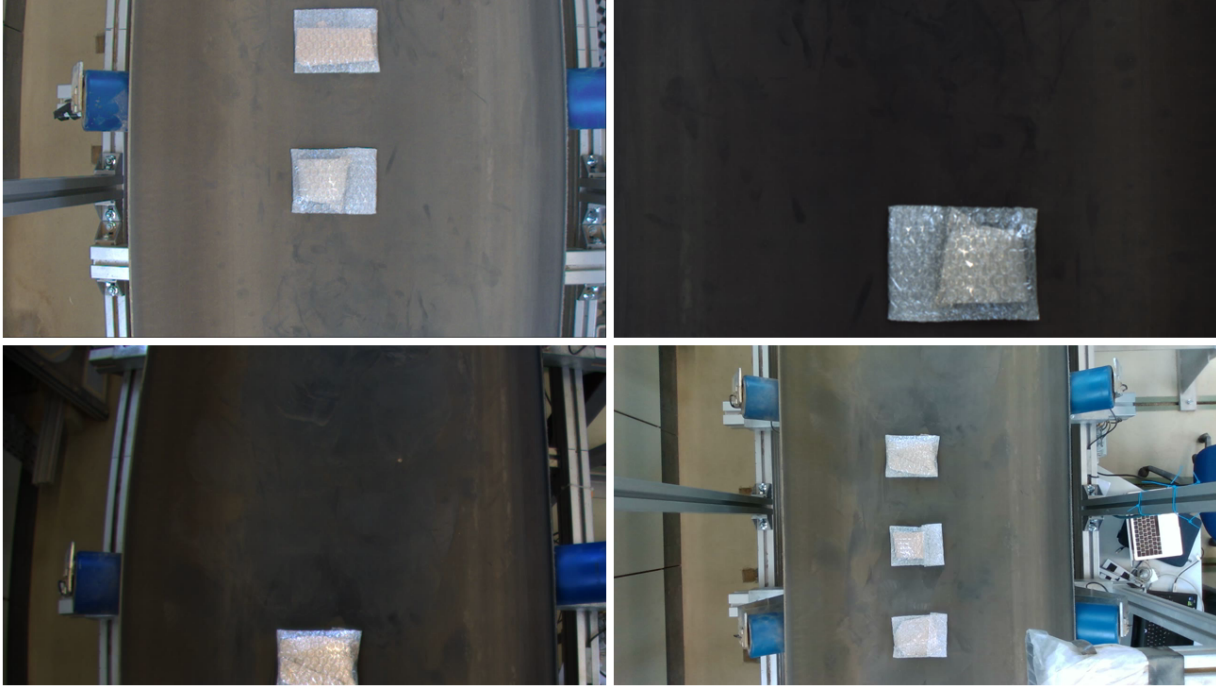


Figure 5: The belt seen by four of the rig cameras (top: basler_1, basler_2; bottom: lucid, rgbd_1), from the timestamped static recording.

The obstacle is that the textbook extrinsic calibration [16, 5] assumes the same board is visible in several cameras at once, which this rig cannot provide: the cameras point at different stretches of belt with little shared view, and basler_1 is an extreme close-up. The response, detailed below, is a board-free registration to a surveyed metric map, which removes exactly that assumption.

5.2.2 Localizing objects: homography versus the alternatives

Turning a detected bag into a belt position is a camera-to-world localization problem, for which multiple view geometry offers several standard routes [6]. Each was weighed against the rig, and the outcome motivates the approach that follows.

Multi-view triangulation reconstructs a full three-dimensional point from two or more cameras that see it at the same instant with known relative pose [6]. It is the most general method, but it needs overlapping and synchronised views. The moving footage here is unsynchronised and the fields of view barely overlap, so triangulation is not available.

Perspective-n-Point (PnP) recovers a camera's full pose from known three-dimensional to two-dimensional correspondences [9] and then back-projects each detection. It yields metric three-dimensional positions but is sensitive to intrinsic quality and to a good spatial spread of correspondences, which lucid and the RealSense cameras could not supply.

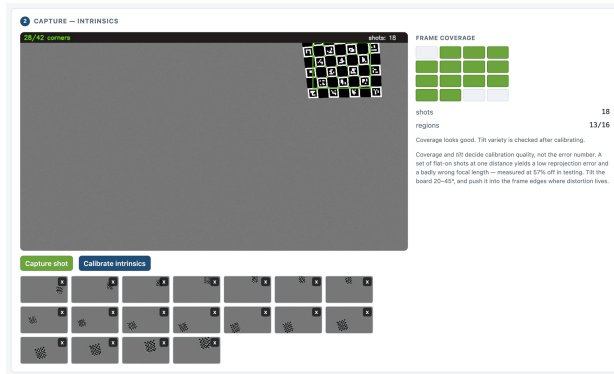
Depth sensing from the RealSense cameras could give position directly, but their intrinsics were unreliable and depth is noisy on glossy, semi-transparent polybags, so it was not depended on.

A *planar homography to the ground plane* exploits the one fact that always holds here: every bag rests on the flat belt. A single homography then maps each camera’s image onto the belt plane [6, 8], needing only four or more coplanar correspondences per camera (fitted robustly with RANSAC [4]), independent of the other cameras and of synchronisation.

The planar homography was therefore chosen, onto one shared surveyed belt map. It is upgraded to a full metric pose where the calibration supports it (the two Basler cameras) and left homography-only where it does not (lucid, RealSense). The choice is dictated by the rig: the belt is planar, the cameras are neither synchronised nor strongly overlapping, and not all intrinsics are trustworthy, so the assumptions behind triangulation and full PnP fail while the single planar assumption holds. The board-free registration below is how each camera’s homography and pose are obtained.

5.2.3 The calibration tool

Calibration uses a purpose-built web application (Figure 6) that runs the whole workflow from recordings, since the camera SDKs were not installed on the development machine. It generates the ChArUco boards, extracts pose-diverse board frames, solves intrinsics and extrinsics, and stores one result file per camera plus a rig plan.



(a) Board capture and detection.

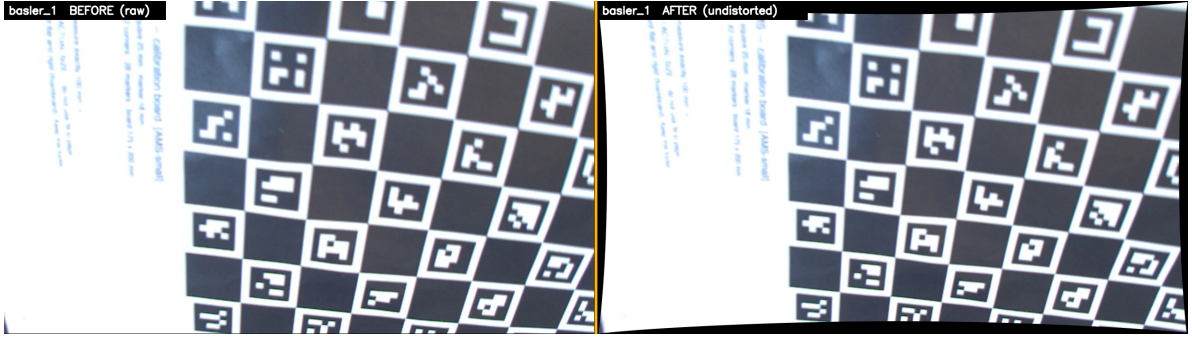


(b) Per-camera intrinsic solve result.

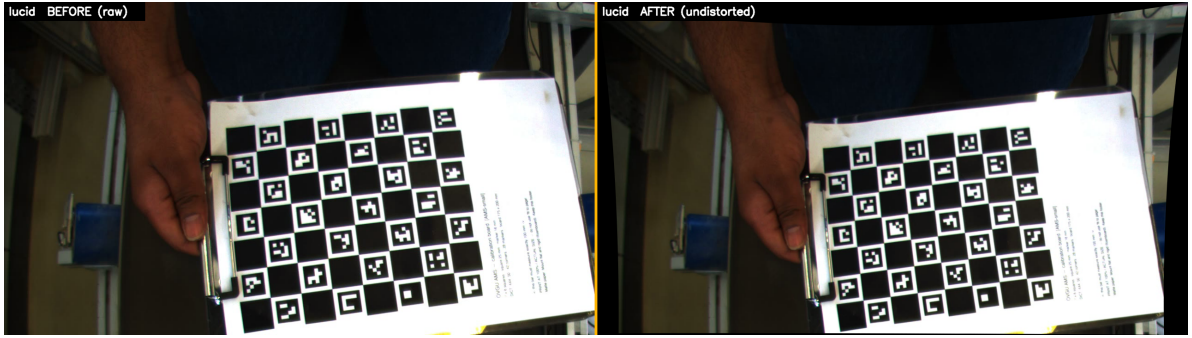
Figure 6: The calibration tool: capture and board detection, and the per-camera solve.

5.2.4 Intrinsics

Per-camera intrinsics are solved from ChArUco captures and checked by undistortion (Figure 7). A validation lesson: a low reprojection error alone does not prove a good calibration. Two perturbation checks (re-solving with the principal point locked to the image centre, and across distortion models) plus a field-of-view sanity check exposed hidden degeneracies, and are how the lucid and RealSense weaknesses were caught.



(a) basler_1 (trusted reference)



(b) lucid (homography-only salvage)

Figure 7: Undistortion check, before (left) and after (right), per camera.

5.2.5 Board-free extrinsics

Placing every camera in the shared frame is the key contribution of the real track. Rather than a shared board view, a single metric map of the conveyor is built once, and each camera is registered to that map from correspondences on one still frame, independently of the others (Figure 8). This board-free registration decouples the cameras, tolerates non-overlapping views, and lets any camera be re-solved on its own, which is what made a shared metric frame achievable on this rig.

Per-camera quality is stated honestly, since it bounds what the twin can claim (Table 5): basler_1 is clean and fully metric, the trusted reference; basler_2 is metric but weaker, agreeing with basler_1 to within about 11 mm on average (22 mm worst) on the same bags; lucid is homography-only (its across-belt position can differ from metric by up to about 236 mm); rgbd_1 needs the RealSense factory intrinsics; rgbd_2 was not calibrated.

5.2.6 The metric belt map

The shared frame is a top-down metric map of the belt, surveyed by tape measure rather than derived from any one camera (Figure 9). Its origin is the board centre under basler_1, with positive Y along the belt and positive X across it. A clean synthetic belt image at exactly one millimetre per pixel, in the same frame, is used for the live display. Each camera reaches the

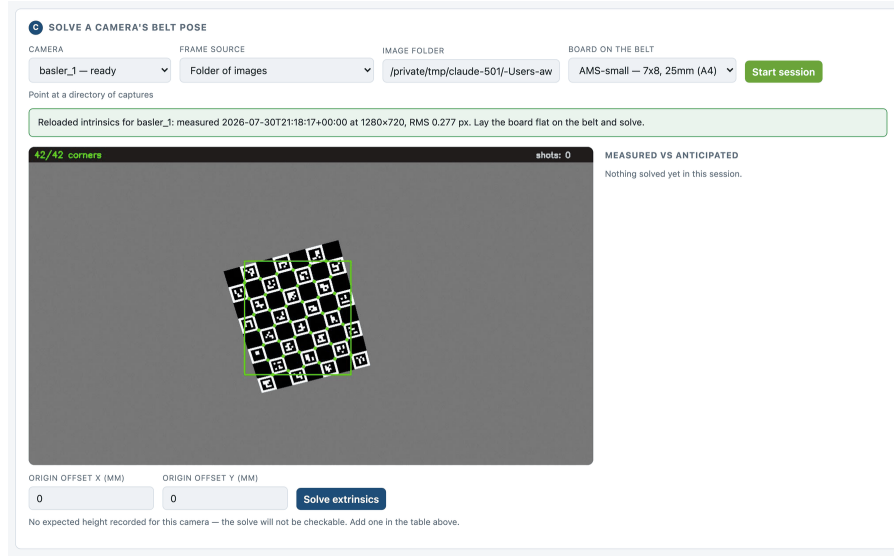


Figure 8: Board-free extrinsic registration: each camera is aligned to the metric belt map independently.

Table 5: Per-camera calibration status.

Camera	Type	Quality / note
basler_1	metric	trusted reference, clean
basler_2	metric	about 11 mm mean, 22 mm worst vs. basler_1
lucid	homography-only	across-belt up to about 236 mm off
rgbd_1_color	homography-only	needs factory intrinsics
rgbd_2_color	not calibrated	

belt through a plane homography that first undistorts the pixel. Because the Basler cameras were calibrated on rectified frames, the live pipeline detects on raw frames and lets this step undistort, reproducing each camera’s stored reprojection error.

5.2.7 Detection and labelling

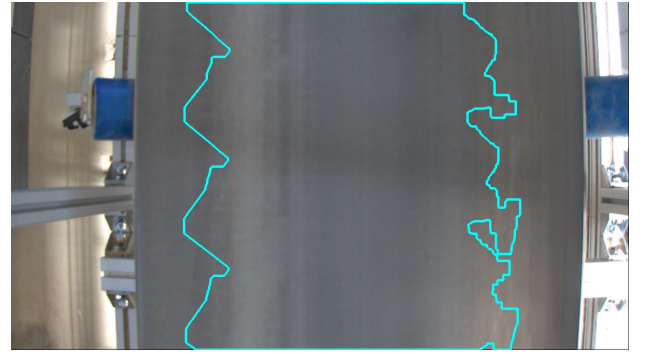
Real bags do not separate by colour, so a real label set was built from scratch. A promptable vision-language localiser (NVIDIA LocateAnything-3B [13]) was LoRA fine-tuned on the supervisor’s annotations and used to auto-label all 2926 frames. The output was cleaned, per-camera belt masks (from temporal variance) suppressed off-belt clutter, and tiled inference [2] recovered dense clusters. The resulting set, labels_final_v3, has 7184 boxes, about 36% more than the untiled set (Figure 10). A cross-check against an independent YOLO11n-OBB detector was instructive: on basler_1 the YOLO reference itself misses more than half the visible bags, so the low agreement there is the reference being wrong, not the labels. The lesson, recurring across the project, is that two models agreeing is not a clean signal when both share a failure mode.



Figure 9: The surveyed metric belt map that defines the shared coordinate frame.



(a) Auto-labelled frame (labels_final_v3, basler_2).



(b) Per-camera belt mask suppressing off-belt clutter.

Figure 10: Auto-labelling of the real footage: LocateAnything boxes after cleanup and tiling, and the belt mask that removes background clutter.

5.2.8 Real detector and benchmarks

For the live system a YOLO11s detector (polybag_v1) was trained on the cluster from labels_final_v3, split by a contiguous per-camera tail to avoid adjacent-frame leakage (Table 6). These scores measure agreement with the auto-label teacher, not accuracy against human ground truth. A separate evaluation against a later hand-annotated set is reported in Section 5.2.11, and confirms the detector is strong on real data. A fine-tuned segmentation model was also trained for mask overlays but has lower recall, so the detect model is the default. Figure 11 shows the training curves and confusion matrix.

Table 6: Real detector (polybag_v1, YOLO11s) on the held-out auto-label split. Scores are agreement with the teacher labels, not human ground truth.

Model	Precision	Recall	mAP50
polybag_v1 (detect)	0.965	0.812	0.864

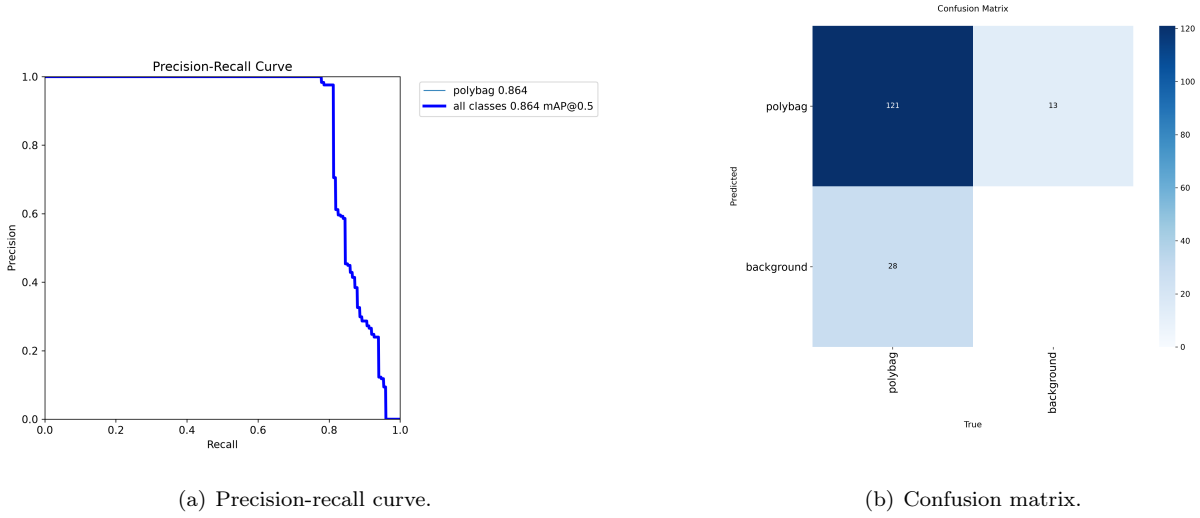


Figure 11: Real detector (polybag_v1) evaluated on the held-out auto-label split.

5.2.9 The live digital twin

The integration deliverable combines detection, calibration, and the map into a live two-dimensional belt map (Figures 12 and 13), reusing the calibration library directly. A streamer runs the detector per frame, converts each detection to belt millimetres, assigns per-camera identities, and publishes all positions over a lightweight MQTT bus. A decoupled browser dashboard draws every bag on the clean map. Each position carries a metric flag (true only for the two Basler cameras) so the display can signal lower confidence, and the projected reference point is the box centre, steadier for flat bags. With per-frame device timestamps (the static set) the streamer achieves true sync (a few tens of milliseconds). The moving sets, having no timestamps, use a normalised timeline. No recording is both moving and timestamped, which is the main data gap. Three scenarios were demonstrated (Figure 14): single moving polybags, dense bulk moving polybags, and a static four-camera capture with true synchronisation. On the laptop the detector runs about five steps per second over three cameras at input size 512, while positions republish at a steady 6 Hz to keep the map smooth.

5.2.10 Fusion and counting

Fusion turns the per-camera detections (already in belt millimetres) into one set of world objects (Figure 15). Overlapping cameras are merged into one object, but only across different cameras,

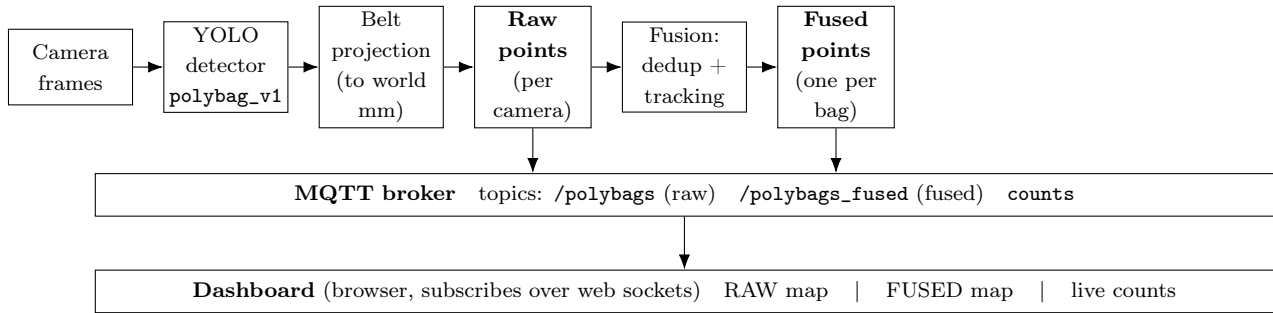


Figure 12: The digital-twin pipeline. Each camera frame is run through the detector and projected onto the metric belt map, giving the per-camera raw points, which fusion deduplicates across cameras and tracks into fused points (one per bag). The producer publishes raw points, fused points, and the counts over MQTT, and the browser dashboard subscribes and draws the RAW and FUSED maps.

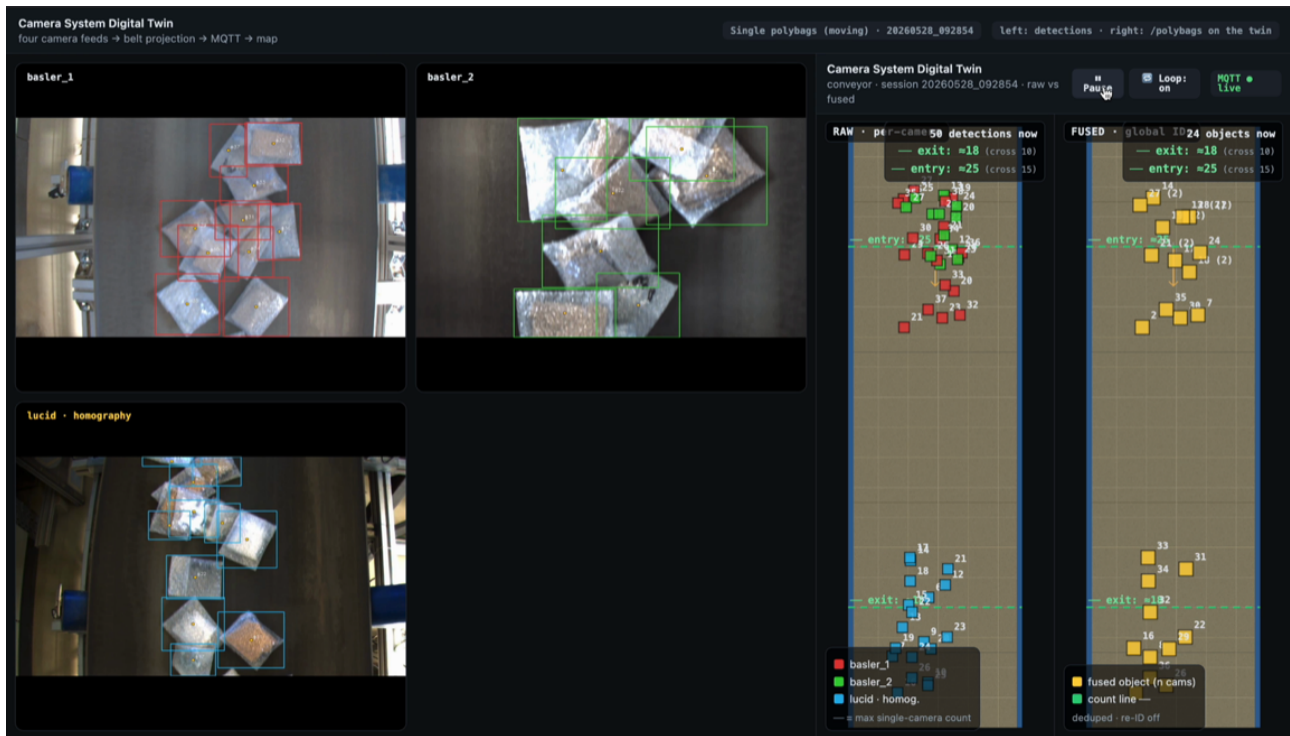
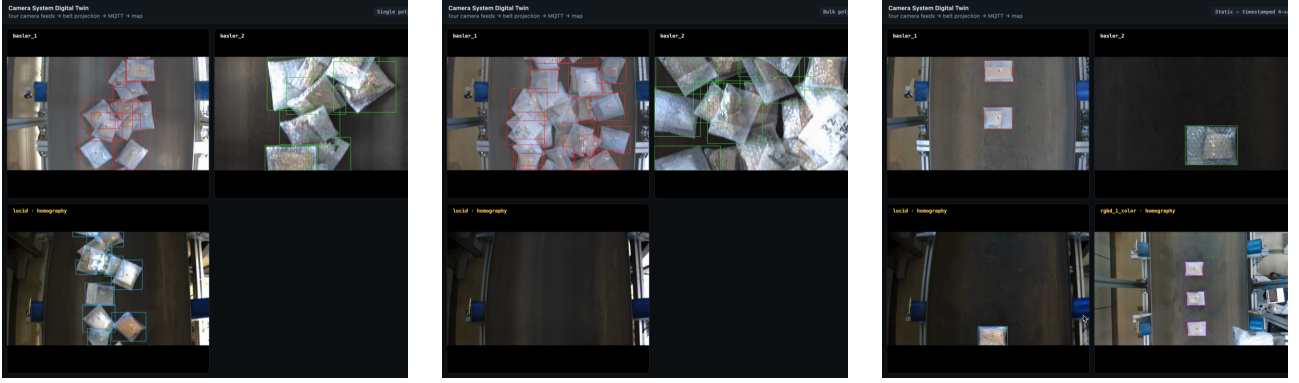


Figure 13: The live digital twin (single moving scenario). Left: the three camera feeds with per-camera detection boxes. Right: the RAW map (one dot per detection, coloured by camera) beside the FUSED map (one object per bag after cross-camera deduplication), with the green entry and exit counting lines.

never within one, since the per-camera tracker has already separated those. A persistent identity is then kept by greedy nearest-neighbour association on the belt plane. Carrying one identity across the blind gap in the middle of the belt (the Baslers see the entry, lucid the exit, nothing the stretch between) is a genuine re-identification problem. Two versions were built and both under-performed, because at a belt speed of about 440 mm per second the roughly 764 mm



(a) Single moving (three cameras).

(b) Bulk moving, dense (three cameras).

(c) Static, four cameras, true sync.

Figure 14: The three demonstrated scenarios, shown by their camera feeds. The moving scenarios (a, b) use the three calibrated moving-set cameras; the bulk scenario is much denser. The static scenario (c) is the only timestamped recording, with all four colour cameras, including `rgbd_1_color`, played in true synchrony, but its bags are nearly static.

gap is crossed in about 1.7 s, and a departure recognised only after a track goes silent fires too late. Cross-gap re-identification was therefore scoped and deliberately deferred, not solved.

Instead, the system counts bags at fixed belt lines, an entry line near the Basler views and an exit line near lucid. Counting each bag by tracking it across the line turned out to be unreliable, because the belt is dense (up to 12 to 21 bags per frame) and fast (about 440 mm per second). At the inference rate a bag travels about as far between frames as the spacing between neighbouring bags, so per-track association merges or fragments them, and the crossing count falls far below the true number (about 10 counted against a manual count near 46). Its apparent stability across replay rates of 5, 10, and 15 Hz was a sign of stably missing the same bags, not of correctness. The count is therefore taken by occupancy flux: the detection occupancy in a thin band around each line is integrated over time and divided by the dwell time of one bag (the band thickness over the belt speed), which estimates throughput without needing a persistent identity per bag. On the bulk clip this recovers about 48, close to the manual count. A negative result supports the design. A ByteTrack tracker did worse here, because fast motion at a low frame rate breaks its overlap association. The simple greedy belt-plane tracker is therefore the right tool, and detector recall, not a fancier tracker, is the lever for a more accurate count.

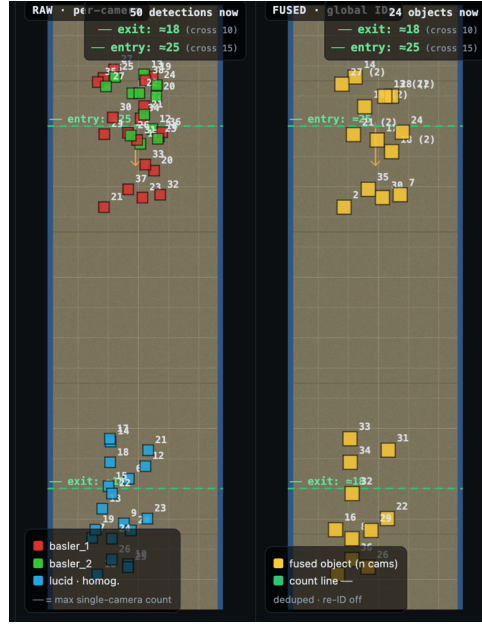


Figure 15: Fusion and counting at a dense moment of the single moving scenario. The RAW map (left, one dot per camera detection, here 50) is deduplicated across cameras into the FUSED map (right, one object per bag, here 24). The green dashed lines are the entry and exit counting lines, each labelled with its occupancy-flux count. A per-track crossing count (shown small in grey) collapses on flow this dense, which is why the throughput count is taken by occupancy flux.

5.2.11 Evaluation on hand-annotated real data

Late in the project the supervisor provided a hand-annotated validation recording (single moving polybags, session of 2 June 2026, 1280 by 720, five cameras, about 50 tracked bags per camera), with oriented polygons carrying a track identity per bag. This is the first human ground truth for the real track. Because the two Basler cameras are the ones the detector was trained on, it gives the in-domain accuracy figure that the auto-label agreement of Table 6 could not. It also lets the same metric families as the synthetic track be reported on real data: detection mAP, and the tracking measures MOTA, IDF1, and HOTA [10] with its detection (DetA) and association (AssA) components.

Detection transfers strongly (Table 7). On the two in-domain Basler cameras the detector reaches a mean mAP50 of 0.963 at a recall of 0.979, close to the 0.995 obtained on synthetic data, and far above a cross-domain check on an unrelated low-resolution RealSense set (mAP50 0.36, resolved once the resolution matched). One RealSense colour view holds up well (rgbd_1, mAP50 0.908) while the other is dragged down by false positives (rgbd_2). The earlier concern that the detector’s real quality was unknown against humans is thus resolved: on the cameras it runs on, it is essentially as accurate as on the simulation.

Tracking tells a more nuanced story (Table 8). The deployed greedy tracker reaches a HOTA of about 65 to 67 on the three well-behaved cameras, and the HOTA decomposition is the informative part: the detection term DetA is high (0.74 to 0.82) while the association term

AssA is the limiter (0.56 to 0.61). Detection is solved, and association, the linking of detections into consistent tracks, is where the tracker loses points, through occasional identity breaks on the fast, low-frame-rate motion. The two standard trackers ByteTrack and BoT-SORT were far worse here (MOTA below 0.1), because at five frames per second their overlap-based association cannot follow the motion. That is precisely why the pipeline uses the simpler greedy spatial tracker. Two independent evaluators (the motmetrics library and TrackEval) agree on MOTA and IDF1, so the figures are trustworthy. Relative to the synthetic track, where MOTA reaches the high nineties, the real numbers are markedly lower, as expected when a single visual class, real false positives, and a low frame rate replace clean simulated data.

The end-to-end count was also checked against this ground truth. On the view whose travel direction matches the pipeline’s convention (rgbd_1), the occupancy-flux count recovers about 48 bags against a ground truth of 50, so the full chain of detection, projection, tracking, and counting is validated where its assumptions hold. On this particular recording the Basler cameras happen to run in the opposite travel direction, which the direction-aware counter does not yet handle, a limitation now understood and straightforward to fix by counting crossings in both directions.

Table 7: Real detector (polybag_v1) against the hand-annotated human ground truth, per camera. The Basler cameras are in-domain (the detector’s training cameras); the RealSense colour cameras are a different sensor at the same resolution.

Camera	Precision	Recall	mAP50	mAP(0.5:0.95)
basler_1 (in-domain)	0.915	0.986	0.985	0.901
basler_2 (in-domain)	0.849	0.973	0.941	0.790
rgbd_1_color	0.867	0.971	0.908	0.707
rgbd_2_color	0.588	0.983	0.583	0.463
Basler mean (in-domain)	0.882	0.979	0.963	0.845

Table 8: Tracking on the hand-annotated human ground truth, per camera, for the deployed greedy tracker. HOTA with its detection (DetA) and association (AssA) components, and MOTA and IDF1 for comparison with the synthetic track. Higher is better; values are percentages.

Camera	HOTA	DetA	AssA	MOTA	IDF1
basler_1	66.9	81.5	55.8	55.1	56.0
basler_2	65.0	73.8	57.6	56.4	71.4
rgbd_1_color	66.8	73.8	60.7	61.9	73.2
rgbd_2_color	55.8	52.5	59.5	7.9	58.4

6 Comparison of Project Plan and Execution

The synthetic track followed the plan closely and delivered the intermediate presentation on schedule on 2 July 2026. The real track, planned as a straightforward transfer of the pipeline, in practice absorbed most of the second half of the project and required several rounds of replanning, as is normal when moving from a clean simulation to a physical rig.

An early real-labelling approach based on watershed segmentation was built for an initial real dataset, but that dataset was retired (its source images were lost), and the approach was retired with it. The real labelling was then rebuilt on the supervisor's new footage with the LocateAnything pipeline described above, which handles the single merged class and the cluttered background far better than a colour-and-morphology method could.

Several concrete defects cost real time and are worth recording. The two Basler video files were swapped relative to their calibration during recording processing, so each camera's calibration was being applied to the other's video. This was diagnosed by cross-camera agreement (the correct assignment brought the two Basler cameras from about 108 mm of disagreement on the same bags down to about 10 mm) and fixed permanently by renaming the files. In the digital twin, binding the message broker to localhost bound only IPv6 on macOS, so the browser's IPv4 websocket was silently refused. Binding explicitly to 127.0.0.1 fixed it. In the fusion stage, an early deduplication gate of 130 mm merged two different bags riding adjacent lanes only about 105 mm apart, which caused a systematic under-count until the rule was changed to merge only across cameras.

Two limitations forced genuine replanning rather than mere debugging. Development ran on a single laptop, whose throughput of about five inference steps per second constrained the input size and the choice of tracker. And the recordings do not cover the case that a synchronised moving demonstration needs: the moving footage has no timestamps and the only timestamped footage is essentially static. The most ambitious planned feature, carrying one identity for each bag across the blind gap in the belt, was scoped, attempted in two forms, and then deliberately deferred once instrumentation showed the recordings could not support it reliably. A robust line count was delivered in its place.

7 Usability of the Results

The project leaves several results that are directly usable. The synthetic track is a reusable benchmarking harness: it generates its own exact ground truth and can evaluate any detector or tracker on standard metrics, which makes it a controlled environment for future method development. The real track leaves a trained polybag detector, a calibrated five-camera rig with a shared metric belt frame, and a working live digital twin that places detected bags on the belt in millimetres in real time over a message bus. Because the dashboard is decoupled

from the producer, the same display can be driven by any future pipeline that emits the same messages. The validated line count is a usable throughput figure for the demonstration clips.

The limitations should be stated plainly. Only the two Basler cameras are fully metric. The lucid and RealSense cameras are homography-only or uncalibrated, so their across-belt positions are approximate, and lucid's offset in particular is the main obstacle to cross-gap re-identification. The detector's training-time numbers are agreement with auto-labels rather than human accuracy. A hand-annotated validation set (Section 5.2.11) has since shown it is in fact strong on the cameras it runs on (mean mAP50 0.963), while tracking is limited more by association than by detection (HOTA about 66, with a high DetA and a lower AssA). Cross-gap re-identification is deferred, not solved. The count sidesteps it by counting at a fixed line. Real-time throughput is limited by a single laptop, and the count is ultimately capped by detection quality. It is taken by occupancy flux, which estimates about 48 bags for the bulk clip (close to a manual count near 46), whereas a per-track crossing count collapses to about 10 on flow this dense, so occupancy flux, not per-track counting, is what makes the figure trustworthy.

8 Conclusions

The project set out to track deformable polybags across several cameras on a conveyor, first in simulation and then on a real rig, and it produced a working pipeline on both. A few lessons stand out. Generating ground truth by projection from known geometry is worth its setup cost: it gave the synthetic track a clean, repeatable benchmark that no amount of manual labelling could match, and it made the detection results on that data trustworthy. Calibrating every camera into one shared metric frame replaced the fragile heuristic association cues (colour and frame order) that could not survive the move to real data, and turned association into a question of distance in millimetres. Honesty in validation repeatedly paid off: perturbation checks exposed calibrations that a low reprojection error would have hidden, a cross-model comparison had to be read carefully because both models shared a failure mode, and a negative result (a more sophisticated tracker did not help at this frame rate) was as informative as a positive one. The right tool depended on the regime, and the simplest method was often the correct one.

A hand-annotated validation set provided late by the supervisor let the real pipeline be measured against human ground truth for the first time, and it confirmed the central claim of the real track. The detector transfers to real data (mean mAP50 0.963 on the in-domain cameras, close to the simulation), and the remaining tracking gap is one of association rather than detection: HOTA of about 66 decomposes into a high detection term and a lower association term. The strong detection and calibration therefore stand on real evidence, and the clearest place to improve is the linking of detections over time.

Several directions remain for future work. On the rig, the RealSense cameras should adopt their factory intrinsics, lucid should be re-recorded with full-frame board coverage for a proper

metric solve, and the belt width should be corrected to 580 mm throughout. A recording that is both moving and timestamped would enable a truly synchronised moving demonstration and is the precondition for revisiting cross-gap re-identification with a proper geometry-driven or appearance-based method. For the throughput count, which is detection-driven, better detector recall remains the lever. For identity-consistent tracking, the human-ground-truth evaluation points instead at association, so a stronger data-association step, a higher frame rate, or making the counter direction-agnostic are where future tracking effort should go. Finally, moving the pipeline off the development laptop and onto hardware at the rig would turn the digital twin from a replay demonstration into a live monitoring tool for the conveyor.

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